



POWER BI MAJOR CAPSTONE PROJECT
END TO END
POWER BI PROJECT GUIDE

PROJECT TITLE	SUPERSTORE SALES & PROFIT ANALYTICS DASHBOARD
PROJECT CATEGORY	Sales Analytics Profit Analytics Business Intelligence
INSTITUTE NAME	Infinity Code Nexus
TRAINER / GUIDE	Anand Kumar Anamaina
CREATED BY	Maheen Begum
TOOLS USED	Power BI Desktop Power Query, DAX & Power BI Service
SUBMISSION DATE	[13/06/2026]

Prepared as part of the Power BI Masterclass Program to demonstrate skills in
Data Analysis, Data Modeling, DAX Development, Dashboard Design, UI/UX,
Storytelling, and Business Intelligence Reporting.

PROJECT OBJECTIVE

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To Develop a comprehensive Power BI dashboard to analyze sales, profit, customers, products, and regional performance for a retail business

- ✓Analyze sales performance
- ✓Evaluate profitability
- ✓Understand customer behavior
- ✓Assess regional performance
- ✓Generate business insights
- ✓Present recommendations

BUSINESS PROBLEM

Retail management requires a clear understanding of sales performance, profitability, customer behavior, and product contribution to make better business decisions.

The objective is to identify:

- Which products generate the highest sales?
- Which regions contribute the most profit?
- Which customer segments drive business growth?
- How discounts impact profit?
- Which products require business attention?

DATASET OVERVIEW

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DATASET NAME : Superstore Dataset

[illegible]

Table 3:DATA

	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	Qu
1	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country	City	State	Postal Code	Region	Product ID	Sales	Qu
2	1	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-BO-10001798	261.96	
3	2	CA-2016-152156	08-11-2016	11-11-2016	Second Class	CG-12520	Claire Gute	Consumer	United States	Henderson	Kentucky	42420	South	FUR-CH-10000454	731.94	
4	3	CA-2016-138688	12-06-2016	16-06-2016	Second Class	DV-13045	Darrin Van Huff	Corporate	United States	Los Angeles	California	90036	West	OFF-LA-10000240	14.62	
5	4	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	FUR-TA-10000577	957.5775	
6	5	US-2015-108966	11-10-2015	18-10-2015	Standard Class	SO-20335	Sean O'Donnell	Consumer	United States	Fort Lauderdale	Florida	33311	South	OFF-ST-10000760	22.368	
7	6	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-FU-10001487	48.86	
8	7	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AR-10002833	7.28	
9	8	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TECH-PH-10002275	907.152	
10	9	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-BI-10003910	18.504	
11	10	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	OFF-AP-100012892	114.9	
12	11	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	FUR-TA-10001539	1706.184	
13	12	CA-2014-115812	09-06-2014	14-06-2014	Standard Class	BH-11710	Brosina Hoffman	Consumer	United States	Los Angeles	California	90032	West	TECH-PH-10002033	911.424	
14	13	CA-2017-114412	15-04-2017	20-04-2017	Standard Class	AA-10480	Andrew Allen	Consumer	United States	Concord	North Carolina	28027	South	OFF-PA-10002365	15.552	
15	14	US-2016-161389	05-12-2016	10-12-2016	Standard Class	IM-15070	Irene Maddox	Consumer	United States	Seattle	Washington	98103	West	OFF-BI-10003656	407.976	
16	15	US-2015-118983	22-11-2015	26-11-2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-AP-10002311	68.81	
17	16	US-2015-118983	22-11-2015	26-11-2015	Standard Class	HP-14815	Harold Pawlan	Home Office	United States	Fort Worth	Texas	76106	Central	OFF-BI-10000756	2.544	
18	17	CA-2014-105893	11-11-2014	18-11-2014	Standard Class	PK-19075	Pete Kriz	Consumer	United States	Madison	Wisconsin	53711	Central	OFF-ST-10004186	665.88	
19	18	CA-2014-167164	13-05-2014	15-05-2014	Second Class	AG-10270	Alejandro Grove	Consumer	United States	West Jordan	Utah	84084	West	OFF-ST-10000107	55.5	
20	19	CA-2014-143336	27-08-2014	01-09-2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-AR-10003056	8.56	
21	20	CA-2014-143336	27-08-2014	01-09-2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	TECH-PH-10001949	213.48	
22	21	CA-2014-143336	27-08-2014	01-09-2014	Second Class	ZD-21925	Zuschuss Donatelli	Consumer	United States	San Francisco	California	94109	West	OFF-BI-10002215	22.72	
23	22	CA-2016-137330	09-12-2016	13-12-2016	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-10000246	19.46	
24	23	CA-2016-137330	09-12-2016	13-12-2016	Standard Class	KB-16585	Ken Black	Corporate	United States	Fremont	Nebraska	68025	Central	OFF-AR-10001492	60.34	
25	24	US-2017-156909	16-07-2017	18-07-2017	Standard Class	SF-20065	Sandra Flanagan	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-CH-10002774	71.372	
26	25	CA-2015-106320	25-09-2015	30-09-2015	Standard Class	EB-13870	Emily Burns	Consumer	United States	Orem	Utah	84057	West	FUR-TA-10000577	1044.63	
27	26	CA-2016-121755	16-01-2016	20-01-2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	OFF-BI-10001634	11.648	
28	27	CA-2016-121755	16-01-2016	20-01-2016	Second Class	EH-13945	Eric Hoffmann	Consumer	United States	Los Angeles	California	90049	West	TECH-AC-10003027	90.57	
29	28	US-2015-150630	17-09-2015	21-09-2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-BO-10004834	3083.43	
30	29	US-2015-150630	17-09-2015	21-09-2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	OFF-BI-10000474	9.618	
31	30	US-2015-150630	17-09-2015	21-09-2015	Standard Class	TB-21520	Tracy Blumstein	Consumer	United States	Philadelphia	Pennsylvania	19140	East	FUR-FU-10004848	124.2	

Data Cleaning Using Power Query

The following transformations were performed:

- Removed duplicate records.
- Corrected data types.
- Handled missing values.
- Removed unnecessary columns.
- Standardized column names.
- Created required calculated columns

Data Model Design

The project follows a structured data model to improve performance.

Fact Table

Sales Fact Table

Contains transactional information:

- Order ID
- Product ID
- Customer ID
- Sales
- Quantity
- Profit
- Discount

Dimension Tables

Customer Dimension

Contains:

- Customer ID
- Customer Name
- Segment

Product Dimension

Contains:

- Product ID
- Category
- Sub-category
- Product Name

Location Dimension

Contains:

- Region
- State
- City

Date Dimension

Contains:

- Year
- Quarter
- Month
- Date

DAX MEASURES CREATED

Total Sales

Calculates total revenue generated.

```
Total Sales = SUM(Sales[Sales])
```

Total Profit

Calculates overall profit.

```
Total Profit = SUM(Sales[Profit])
```

Total Orders

Counts the number of orders.

```
Total Orders = DISTINCTCOUNT(Sales[Order ID])
```

Profit Margin

Measures profitability percentage.

```
Profit Margin = DIVIDE([Total Profit],[Total Sales])*100
```

Average Order Value

Calculates average sales per order.

```
Average Order Value = DIVIDE([Total Sales],[Total Orders])
```

DASHBOARDS

Executive Overview

Purpose

Provides a summarized view of overall business performance.

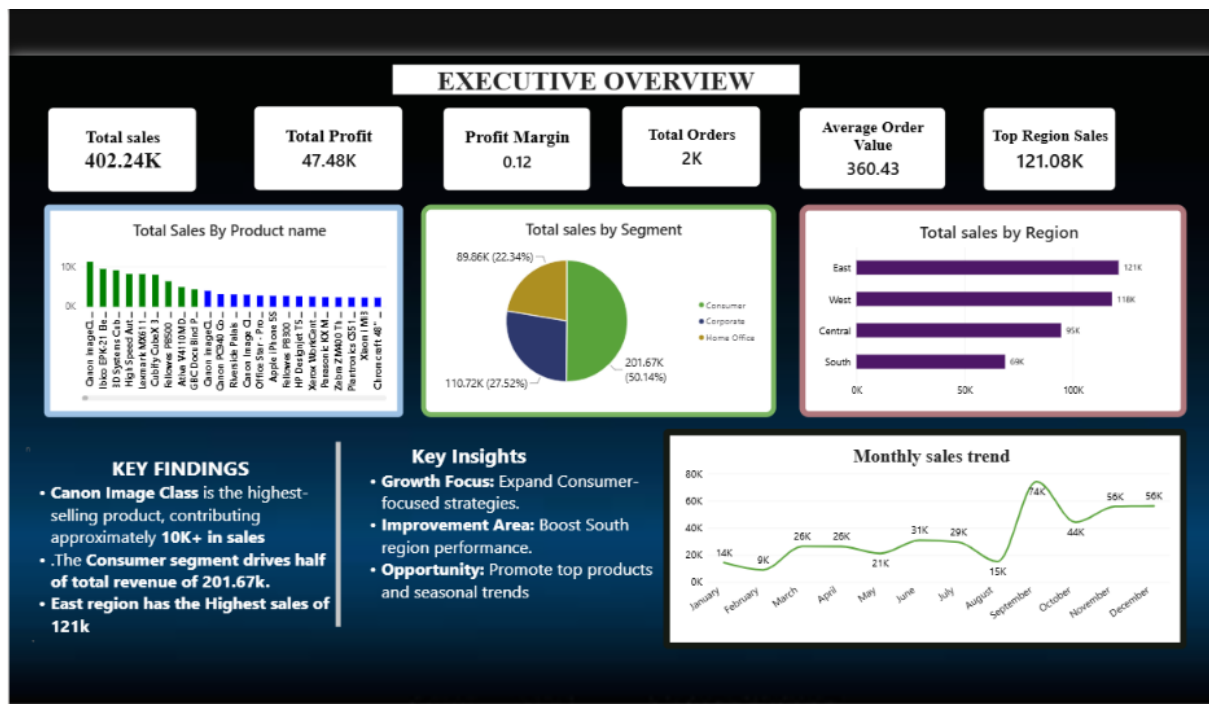
KPIs Created

- Total Sales
- Total Profit
- Profit Margin
- Total Orders
- Average Order Value

Visuals Created

- Sales trend chart
- Regional performance chart
- Customer segment analysis
- Category performance

Business Value: Allows executives to quickly understand business health.



Sales Analysis Dashboard

Purpose

Analyzes sales performance across different dimensions.

Visualizations Created

Monthly Sales Trend

Shows revenue movement over time.

Category Sales Analysis

Displays sales contribution from different categories.

Regional Sales Comparison

Compares sales across geographical areas.

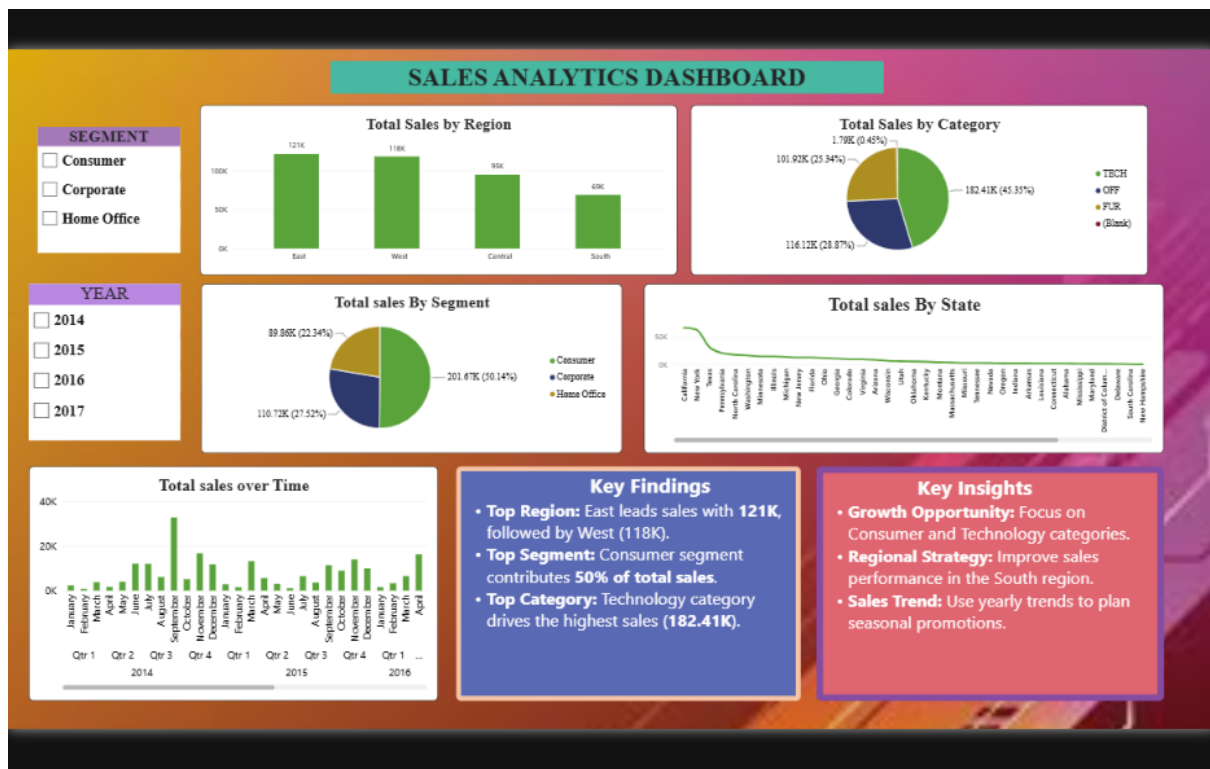
Sales Distribution

Shows contribution percentage by category.

Features

Drill Down:

Year-> Quarter-> Month-> Date



Profit Analysis Dashboard

Purpose

Analyzes profitability performance.

KPIs Created

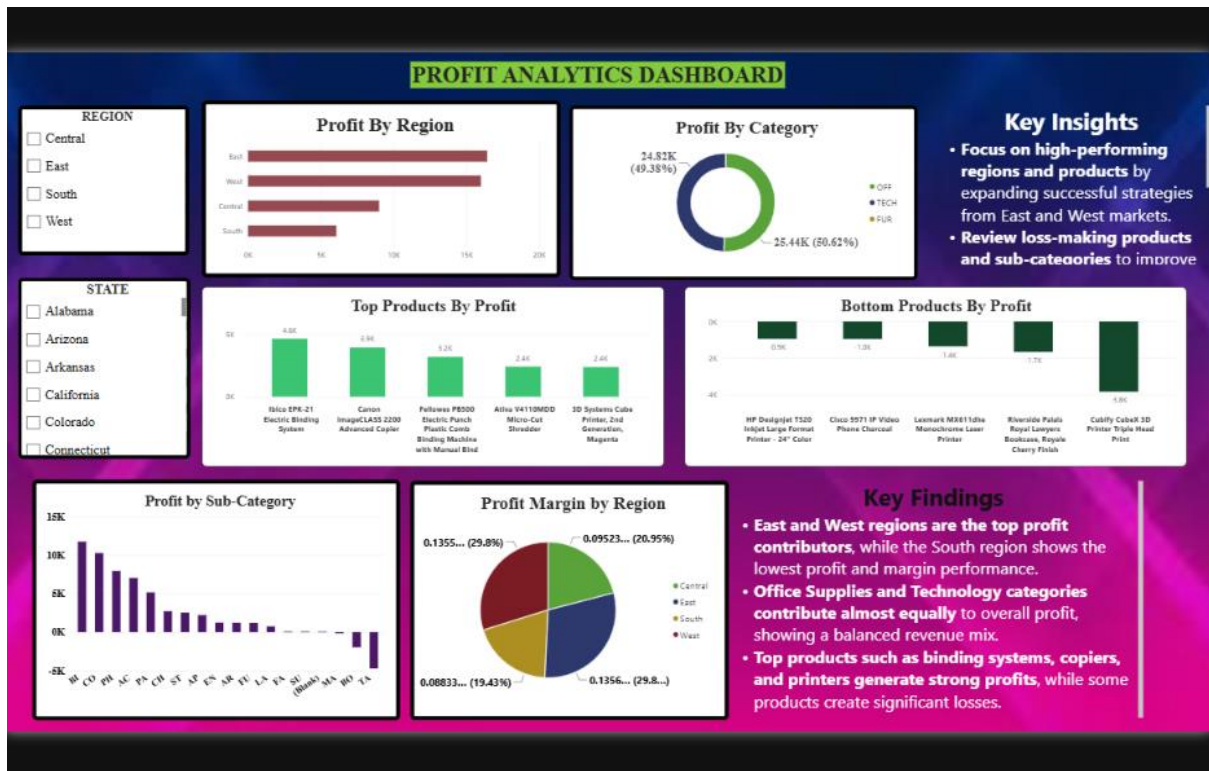
- Total Profit
- Profit Margin
- Most Profitable Products
- Least Profitable Products

Visualizations created

- Profit by category
- Profit trend
- Profit by region
- Discount impact analysis

Business Value

Helps improve profitability by identifying profitable areas.



Customer Analysis Dashboard

Purpose

Understanding customer behavior and value.

Analysis Areas

- Customer segments
- Customer contribution
- Order frequency
- Revenue generated by customers

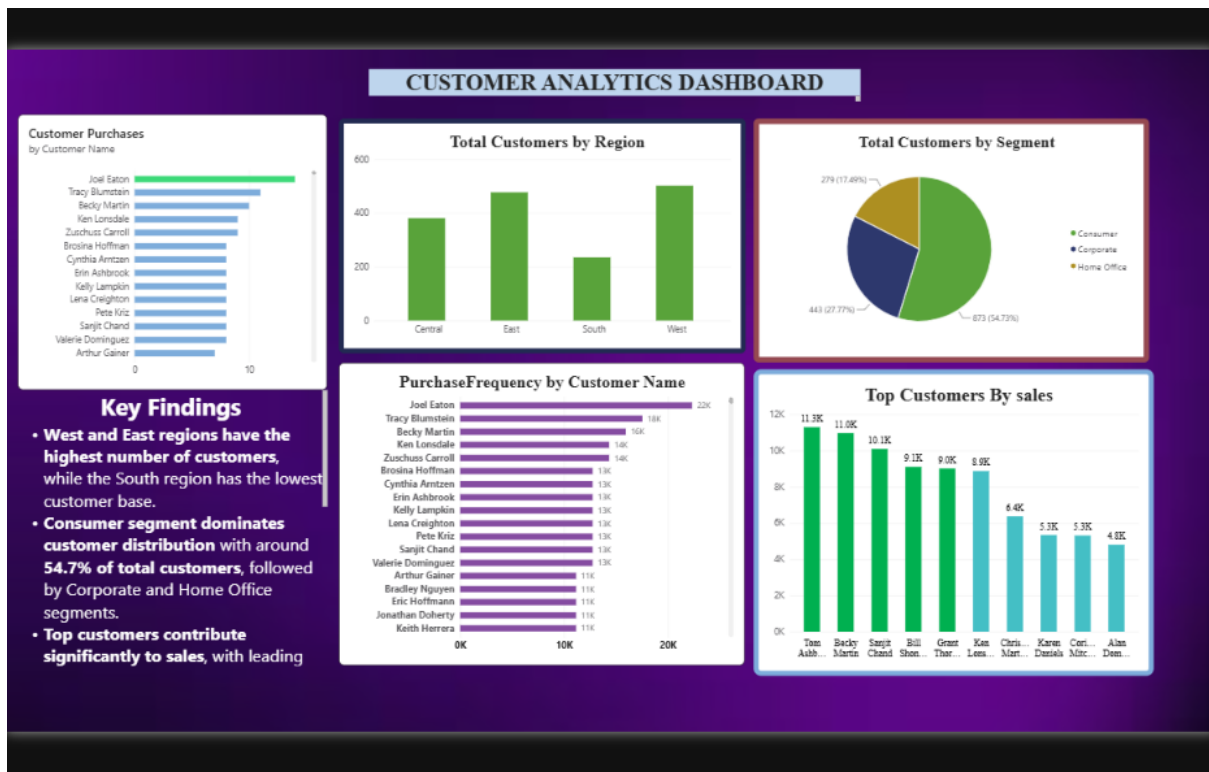
Visuals created

- Sales by segment
- Top customers

Drill Through

Allows navigation from summary reports to detailed pages.

Business Value: Supports customer-focused strategies.



Product Analysis Dashboard

Purpose

Analyzes product performance.

Visualizations created

- Top selling products
- Product category comparison
- Sub-category performance
- Product profitability

Business Value

Helps improve inventory and product decisions.

Dashboard Features

Filters and Slicers

Users can filter data using:

- Year

- Region
- Category
- Customer Segment
- Product

Drill Down

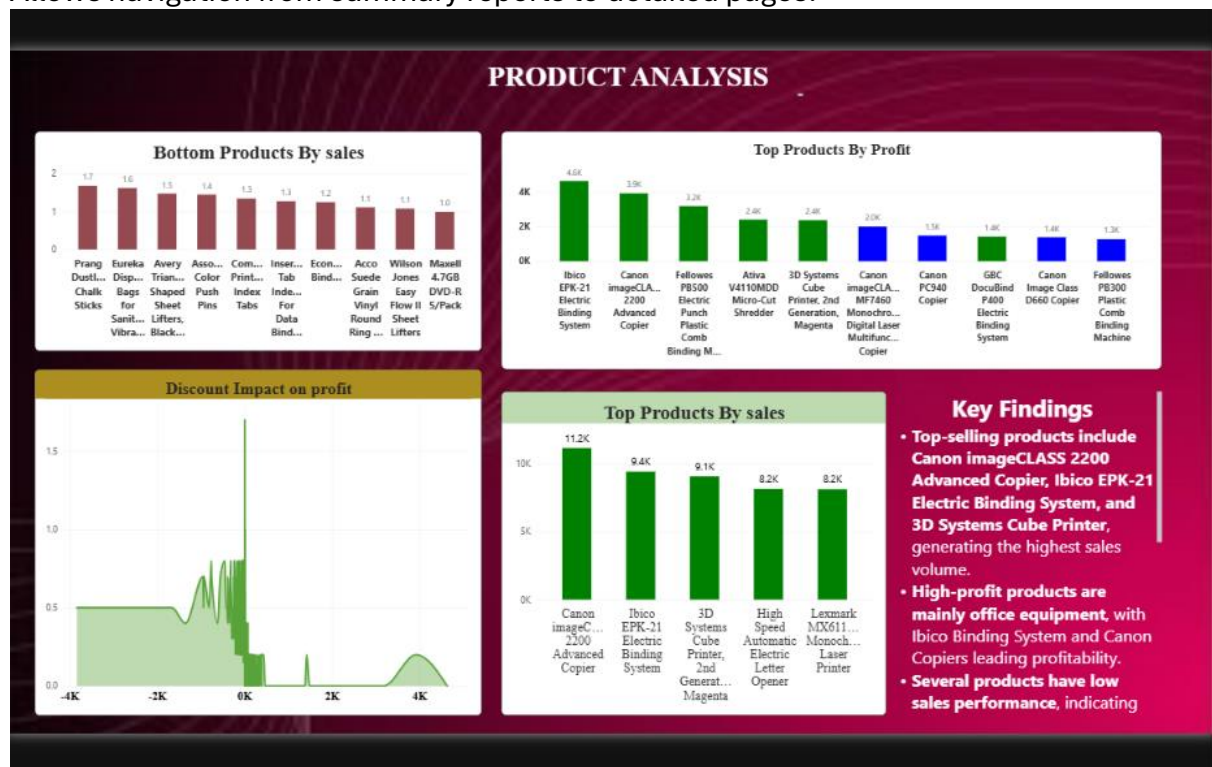
Users can analyze detailed information:

Region → State → City

Category → Sub-category → Product

Drill Through

Allows navigation from summary reports to detailed pages.



Key Business Insights

The dashboard helps identify:

- Revenue performance trends.
- High-performing products.
- Profitable categories.
- Customer purchasing behavior.

- Regional opportunities.
- Areas requiring improvement.

Testing and Validation

Testing performed:

Data Validation

- Compared dashboard values with source data.
- Verified calculations.

Visual Testing

- Checked filters.
 - Tested drill-down functionality.
 - Verified dashboard responsiveness.
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Limitations

- Dashboard accuracy depends on source data quality.
 - Real-time updates require a live data connection.
 - Historical analysis depends on available records.
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Future Enhancements

Possible improvements:

- Real-time sales tracking.
 - Predictive sales forecasting.
 - AI-based recommendations.
 - Automated alerts.
 - Integration with live databases.
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Conclusion

The Sales and Profit Analytics Dashboard developed using Power BI provides a complete Business Intelligence solution for analyzing business performance. The dashboard transforms raw sales data into actionable insights through interactive reports, KPIs, and advanced analytics.

This project helps organizations improve decision-making, optimize sales strategies, increase profitability, and monitor overall business growth effectively.